

SEHH2042 Computer Programming

Individual Assignment 1

Submission deadline: 15 March 2025 (Saturday), 23:59

Expected Learning Outcomes

- Develop computer programs in one or more high level language programming environment.
- Design and develop structured and documented computer programs.
- Integrate the computer programming techniques to solve practical problems.

Introduction

This is an **individual assignment**. There are 2 parts:

Criteria	% of marks
ShowInfo	10
Write a C++ program	90

You are given a C++ program template file called *A1Template.cpp*. You are required to **insert C++ codes into the template file** according to the given instructions. The final file (your submission) will be a complete C++ program, which can be compiled and executed successfully and is able to satisfy all the requirements in this specification.

Instruction

- For each question, you only need to do **either odd or even version** based on the stated rule about your student ID number (0 is considered as even). You do not need to write any program codes to check the student ID number to determine the version you should do.
- To answer the questions, you need to insert codes into the functions of *the template file*. E.g., to answer question 1, write your code in the scope of *Q1()*. When the program is executed, enter the question number to run the code of a particular question.
- In each question, the program should first accept input(s) from user **WITHOUT** printing any prompt messages (Hint: *cin* statements will be used, and there is **NO** output messages like "Input n: " before the *cin* statements).
- You may assume that user always provides valid inputs (correct type and correct range) as required by the question. **NO error input checking** is needed unless otherwise required.
- Follow **EXACTLY** the requirements as stated in the questions. Check the correctness of your code by executing it and compare its output with the sample display.
- Test your program carefully by using input test cases as stated in the sample display, as well as other necessary test cases you can think of. Correct output with same output format as the sample display is expected.

- Marks are awarded based on the number of test cases that your program can successfully pass without mistakes.
- You may include more header files and write user-defined functions to solve the questions. E.g., you may write a user-defined function for solving question 1, and call it in the given function `Q1( )`. (*refer to “Using template file” on the last page*)
- Apart from inserting codes as mentioned above, you are **NOT** allowed to modify any given codes in the template file.
- **IMPORTANT:** Make sure that your submitted file **CAN be opened** and has **NO syntax error**. It should be the source code, using .cpp as the file extension. Your submission should **NOT** be other file types such as .zip or .sln.

ShowInfo (10%)

Insert your codes inside the showInfo function block so that your personal particulars are displayed in the following format when the program executes:

Sample display:

```
Name       : XXX YYY ZZZ
Student ID: 24xxxxxxA
Class      : Bxx
```

Question 1 (30%)

Write your codes inside Q1 function block that repeatedly accepts integer inputs from the user, until the user enters 0 that indicates the end of user input. The program must process each integer according to the **4th digit** of your student ID (e.g. 123**4**5678A).

Note that the last input 0 is not counted. You can assume that the users always enter one integer in each round of input.

Odd Version (odd 4th digit in student ID)

Display the following items:

- *Count the integer if it is positive and odd.*
- *Count the integer if it is negative and even.*
- *The sum of the above integers.*

Sample display

3

-4

8

-2

5

-6

0

2 ← Positive and odd: 3, 5

3 ← Negative and even: -4, -2, -6

-4 ← Sum of above: 3-4-2+5-6

Sample display

7

-8

-3

2

-10

1

0

2

2

-10

Even Version (even 4th digit in student ID)

Display the following items:

- *Count the integer if it is positive and even.*
- *Count the integer if it is negative and odd.*
- *The sum of the above integers.*

Sample display

4

-3

7

-5

2

-8

0

2 ← Positive and even: 4, 2

2 ← Negative and odd: -3, -5

-2 ← Sum of above: 4-3-5+2

Sample display

8

-1

-7

-5

6

3

0

2

3

1

IMPORTANT NOTE: Compare your program output with the sample display. There should be **NO** prompt messages and **NO** additional blank lines in the output. **DO NOT add extra text** in the output which will be considered as incorrect in marking.

Question 2 (30%)

Write your codes inside Q2 function block that calculates the total cost for a car charging session according to the table below. The program should accept the following user inputs **in order**. Note that **NO** prompt messages are needed.

- The charger type as either “fast” (f) or “normal” (n) (a *char* value)
- The start time of the charging session (in hour) (an *integer* value)
- The end time of the charging session (in hour) (an *integer* value)

In the calculation, if the total charging fee (before adding the parking fee) exceeds \$100.00, there is a 15% discount on the charging fee. The parking fee is not discounted. After the calculation, the program shows the total cost in **two decimal places**.

The charging fee and parking fee are given by the table below, determined by the 5th **digit** of your student ID (e.g., 1234**5**678A).

	Price /hour (Odd 5 th digit)		Price /hour (Even 5 th digit)	
Charging Fee:	Fast	Normal	Fast	Normal
Peak Hours (08:00-18:00)	\$60	\$20	\$70	\$22
Non-Peak (18:00-22:00)	\$45	\$15	\$50	\$16
Overnight (22:00-08:00)	\$8	\$5	\$10	\$6
Parking Fee: • For fast charging , first 1 parking hour is free. • For normal charging , first 4 parking hours are free.	\$20		\$22	

Sample display (for odd)

f ← Charger type (f or n)
8 ← Start time (hour)
9 ← End time (hour)
60.00

Sample display (for odd)

f
17
20
167.50

Sample display (for odd)

n
20
3
115.00

Sample display (for odd)

n
7
13
129.25

Sample display (for even)

f
8
9
70.00

Sample display (for even)

f
17
20
188.50

Sample display (for even)

n
20
3
128.00

Sample display (for even)

n
7
13
142.60

IMPORTANT NOTE: Compare your program output with the sample display. There should be **NO** prompt messages and **NO** additional blank lines in the output. **DO NOT add extra text** in the output which will be considered as incorrect in marking.

Question 3 (30%)

Write your codes inside Q3 function block that displays the pattern (letter 'p' or 'q') as shown below. The user input is an integer which determines the number of rows and columns in the pattern.

If the input integer is 0, negative or an even number, then your program should print the character 'E' (which represents error).

Determine the version of your program by the **6th digit** of your student ID (e.g., 12345678A).

Odd Version (odd 6th digit in student ID)

Sample display

```
7
*****
*       *
*       *
*****
*
*
*
```

Even Version (even 6th digit in student ID)

Sample display

```
7
*****
*       *
*       *
*****
*
*
*
```

More examples

Input	Output (Odd Version)	Output (Even Version)
5	***** * * ***** * *	***** * * ***** * *
3	*** *** *	*** *** *
1	*	*
4	E	E

IMPORTANT NOTE: Compare your program output with the sample display. There should be **NO** prompt messages for the input size and **NO** additional blank lines in the output. **DO NOT add extra text** in the output which will be considered as incorrect in marking.

Submission

You are required to **insert your C++ code into the given template file**, and submit the **final** source file to Blackboard (centralized page) before the deadline. Use your name and student ID as the filename: ***StudentID_Name.cpp***. Remove all spaces, hyphens and other non-letter characters in the filename. A correct filename should look like: 12345678A_ChanTaiMan.cpp.

Grading

Your program (i.e. the template file with your answers) will be executed by script with different test cases in **Microsoft Visual Studio** using the **Release** setting. The tester will execute the program and enter the question number in “Program Selection Menu” to test a particular question. The program will be restarted for testing each question individually.

You need to follow **EXACTLY** the above input and output requirements. Any deviation from the requirement is considered as incorrect and **no mark** is given for that test case.

Marks deduction

Late submission: 100% deduction. **No late submission is allowed.** Submit your work to Blackboard some time ahead of the deadline. Late submissions due to slow internet speed will not be accepted.

Syntax error: 5% - 20% deduction depends on the seriousness of the syntax error. You will get **0 mark** if your program contains too many syntax errors. Check your final source file using Microsoft Visual Studio (not those online compilers) carefully before submission.

Runtime error: No mark for the particular test case that triggers the runtime error (e.g. infinite loop, divide by zero, etc.).

Logic error (bug): No mark for the particular test case that deviates from the requirement. Note that a logic error may lead to **failure in ALL test cases** of a question, e.g. displaying incorrect messages, incorrect spelling and spacing, incorrect number format, or incorrectly decide the odd/even version, etc.

Unnecessary prompt messages: 15% deduction.

Ensure the originality of your work. Plagiarism in any form is highly prohibited.

Checklist for your use:

- ☐ My source code has been successfully compiled under Microsoft Visual Studio.
- ☐ My source code has been tested with different test cases, including my own test cases.
- ☐ There are **NO** prompt messages when my codes in Q1, Q2 and Q3 are executed.
- ☐ My personal particulars have been filled in the showInfo function block.
- ☐ My final submission is in .cpp format with a correct filename.

Additional Information

If you implement the questions in separated source files, you need to copy the program codes into the template file for assignment submission. Make sure to **test the final source file (i.e. template file with your answers)** in Microsoft Visual Studio before submission.

myQuestion1.cpp

```
#include <iostream>
#include <iomanip>
using namespace std;

void display(int n) {
    cout << "This is appendix\n";
    cout << "Display a number: " << setw(5) << n;
}

int main() {
    int number = 1234;
    display(number);
    return 0;
}
```

Template.cpp

```
// Insert more header files when necessary
#include <iostream>
#include <iomanip>
using namespace std;

void showInfo()
{
    // Insert your codes to display your personal particulars here
}

// Insert your function, class (if any) for Q1() here
void display(int n) {
    cout << "This is appendix\n";
    cout << "Display a number: " << setw(5) << n;
}

void Q1()
{
    // Insert your codes for Question 1 here
    int number = 1234;
    display(number);
}

// ... the rest of the template file ...
```

1. The **header files** included in your program should also be included in the template file.
2. The **user-defined function / class** for a question should be copied **before** the question.
3. The program **main body**, except “**return 0**”, should be copied to the function body of the corresponding question.

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